



Part 1 What does energy look like?

The core activities in this unit total about 22.5 hours of lesson time. The Content Descriptions (Australian Curriculum: Science) for each activity are listed here as curriculum codes and described fully in each lesson plan.

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
1.1	Spot the energy	ACSSU155 AC SIS148	Engage & Explore	Students explore a range of phenomena and identify different types and stores of energy.	Core
			Classroom & Digital		
			Medium		
1.2	Energy to burn	ACSSU155 AC SIS148 AC SHE136	Engage & Evaluate	Students watch a video of a rocket launch and identify some of the energy forms and transfers occurring. They will recall words associated with energy from Activity 1.1 and will learn two terms associated with mechanical energy: kinetic and gravitational potential energies.	Core
			Short		
			Medium		
1.3	Forms of energy	ACSSU155 AC SIS148	Explore, Explain & Evaluate	Students will label a magazine picture, identifying all energy related phenomena.	Core
			Classroom		
			Short		

Part 2 Why do things move?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.1	Throw the ball	ACSSU155 ACSIS145 ACSIS148	Explore & Explain	Students will observe someone throwing a ball as high in the air as possible. They will discuss the forces involved while the ball is being thrown and during its flight. They will also identify forms of mechanical energy involved; gravitational potential and kinetic energies, and how a transformation between these two forms of energy allows for the conservation of mechanical energy.	Core
			Classroom & Digital		
			Short		
2.2	Energy skate park	ACSSU155 ACSIS234	Explain & Elaborate	Students will explore the Energy skate park learning object and apply their understanding of kinetic and gravitational potential energies to explain the skater's motion.	Optional
			Digital		
			Medium		
2.3	Energy in, energy out	ACSSU155 ACSIS148	Explore & Explain	Students will explore the concept of work through an investigation of simple levers and pulley systems.	Core
			Hands-on		
			Long		
2.4	Heat from friction	ACSSU155 ACSIS144 ACSIS145 ACSIS234 ACSIS148	Explore	Students will conduct a simple experiment exploring the link between friction and the heating of water.	Core
			Hands-on		
			Medium		
2.5	Starting fires	ACSSU155 ACSHE226 ACSIS148	Explore & Explain	Students learn about the use of fire through human history and indigenous perspectives of the use of fire for human survival.	Optional
			Hands-on		
			Long		
2.6	Putting it together	ACSSU155 ACSHE226 ACSHE136	Elaborate & Evaluate	Students will evaluate their understanding through class discussion and completion of a quiz.	Core
			Classroom		
			Medium		

Part 3 What is heat? What is temperature?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.1	What is heat?	ACSSU155 ACSHE226	Engage & Explain	Students are introduced to a range of phenomena that relate to the concept of heat. They then watch a video about Rumford's discovery that heat was a form of energy that could be generated through friction. Students explore their own ideas and formulate their own hypotheses as they progress through the activity.	Core
			Digital		
			Medium		
3.2	Keeping your drink warm	ACSSU155 ACSIS139 ACSIS140 ACSIS141 ACSIS234 ACSIS148	Engage & Explain	Students explore different ways of keeping a hot drink warm, and relate these to conduction & convection.	Core
			Hands-on		
			Medium		
3.3	Water is hard to heat	ACSSU155 ACSIS139 ACSIS144 ACSIS145 ACSIS234	Explore, Explain & Elaborate	Students explore the relationships between heat, temperature, volume and heat capacity, particularly of water.	Core
			Hands-on		
			Long		
3.4	Feeling hot and cold	ACSSU155 ACSHE136 ACSIS139 ACSIS144 ACSIS234	Explore & Explain	Students will learn about the importance of conduction in how hot or cold surfaces feel.	Core
			Hands-on & Digital		
			Medium		
3.5	Heating water with light	ACSSU155 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Explain	Student will conduct a laboratory experiment to determine whether the colour of a surface affects the heating effect of light.	Core
			Digital		
			Medium		
			Hands-on & Digital		
			Long		

Part 3 What is heat? What is temperature? (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.6	Solar energy	ACSSU155 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS146 ACSIS234	Explore, Explain, Elaborate & Evaluate	Students design and test a solar oven for heating water. They will follow a basic plan to design a solar collector that will collect and focus as much light on a container of water to maximise heating.	Core
3.7	What is sunlight?	ACSSU155 ACSIS144	Explore & Explain Hands-on & Digital Medium	Students will use a light box to visualise the spectrum of light. Notebook questions draw together their ideas about light and colour.	Core
3.8	Cyclones	ACSSU155	Explain & Elaborate Digital Medium	Students explore digital resources and apply knowledge to explain how the sun's energy drives a cyclone.	Optional

Part 4 What is electricity?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.1	Making a globe light up	ACSSU155 ACSIS140 ACSIS144	Engage & Explore Hands-on Medium	Students discover the fundamentals of making an electric circuit work.	Core
4.2	Showing electric circuits	ACSSU155 ACSIS144	Explore Classroom Short	Students will design their own symbols to represent components in an electric circuit.	Core
4.3	Heating with electricity	ACSSU155 ACSIS140 ACSIS141 ACSIS145	Explore & Explain Hands-on Medium	Students construct an electric water heater using high resistance wire.	Optional
4.4	Build a motor	ACSSU155 ACSHE 134 ACSIS140 ACSIS145	Explore & Elaborate Hands-on & Digital Medium	Students will follow instructions to build a model motor.	Core
4.5	Build a generator	ACSSU155 ACSHE 134 ACSHE 135 ACSIS144 ACSIS145 ACSIS234	Explore & Elaborate Hands-on Short	Students build a simple generator using basic equipment.	Core
4.6	The lemon battery	ACSSU155 ACSIS144 ACSIS145	Explore & Explain Hands-on Medium	Students explore the best way to make an LED glow using a lemon battery.	Optional

Part 5 Why do we eat?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.1	Food as fuel	ACSSU155 ACSIS140 ACSIS144 ACSIS145	Explore & Explain	Students will conduct a simplified calorimetric experiment to measure the energy content of a nut	Core
			Hands-on		
			Medium		
5.2	Energy in food	ACSSU155	Explore	Students investigate the energy content of a range of foods and relate this to daily needs. They judge the quality of different foods.	Core
			Classroom & Digital		
			Medium		
5.3	Who eats what?	ACSSU155 ACSIS148	Explain, Elaborate & Evaluate	Students use the Classroom Antarctica website to build an Antarctic food web. They will complete a simple exercise linking food chains and food webs to the transfer of chemical energy through ecosystems.	Core
			Classroom		
			Short		
5.4	Rates of photosynthesis	ACSSU155 ACSIS139 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS148	Explain, Elaborate & Evaluate	Students carry out a digital activity and conduct an experiment to investigate one of the factors affecting photosynthesis.	Core
			Hands-on & Digital		
			Long		

Part 6 How do we energise our cities, towns and homes?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
6.1	Where does our energy come from?	ACSSU155 ACSHE 134 ACSHE 135 ACSHE136	Elaborate & Evaluate	Students investigate, using on-line and local resources, where their electricity supply comes from and how it is generated.	Core
			Digital		
			Medium		
6.2	Energy systems	ACSSU155 ACSIS148	Elaborate	Students examine several systems involving multiple energy transfers and use a Sankey diagram to represent and analyse each (e.g. energy transfer in a car and domestic household).	Core
			Digital		
			Medium		
6.3	Conserving energy	ACSSU155	Explore, Explain, Elaborate & Evaluate	Within the context of their own location, students explore ways of minimising heat transfer in a house to save energy used in heating or cooling. Students will design and evaluate an energy efficient house that minimises the movement of heat into or out of the house.	Optional
			Digital & Classroom		
			Medium		
6.4	Honda ad	ACSSU155 ACSHE136 ACSIS145 ACSIS148	Engage, Explain & Evaluate	Students watch the link to the UK Honda Accord commercial and identify: - energy forms and transfers - energy stores in the system.	Core
			Digital		
			Medium		
6.5	Sample test		Evaluate	A sample summative test (editable version) and marking scheme are available to teachers by email request from <i>Science by Doing</i> (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - <i>Science by Doing</i> provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. <i>Science by Doing</i> encourages teachers to modify the items to suit individual teaching programs.	Optional
			Classroom		
			Long		

1.1 Spot the energy?

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy transfers involve change.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- recognise when energy is being transformed in different situations
- identify different energy stores at the beginning and end of processes and events.

What ideas might your students already have?

- students may not be familiar with identifying energy in everyday situations.
- students may tend to associate energy only with living things.
- students may tend to associate energy only with movement.
- students may only equate energy with fuel.

Key vocabulary:

Energy, kinetic energy, potential energy, heat, temperature, conservation

Equipment list

Each CLASS will require:

Eight stations comprising a learning centre with different energy transfer phenomena. These are most effective if they are simple.

They might include: simple wind-up and battery-operated toys; candle; torch; wind-up torch; ball rolling down a slope; simple pendulum; melting ice block; spring on hook; drum; toy guitar; baking powder, weak acid (vinegar) and test tube. Build a nice collection that can be used and added to each year.

Each station should be labelled so students can identify it in their **Notebook**.

Each STUDENT will require:

- access to *Student Digital Activity 1.1*
- **Notebook**

Things to consider

You will need a plan to ensure groups progress around the activities. If you have more activities than groups, this can be self-paced, although check groups don't get stuck on an activity. If the class is large, you may need time limits so all groups move on at the same time.



Ensure each form of energy is represented in at least one station. These should include: sound, light, kinetic energy, potential energy, chemical, electrical, heat.

In this first activity students may not use all terms correctly. They may use the word *gravity* to explain why a ball has the energy to roll down a slope. This can be accepted at this stage as a general idea and refined in later activities. This may be true for other words students offer.

Teacher content information

In identifying energy we often refer to different energy forms. This can imply that energy is a material substance. We can also refer to energy stores. For example, a moving object has energy stored as movement.

Key Ideas:

- Energy is involved when changes occur or things happen.
- Energy can be stored in different ways.

Lesson plan

As you teach:

Step 1: Students undertake *Student Digital Activity 1.1*, which provides a simple introduction to different energy forms and transformations. This could be a simple homework exercise.

Step 2: Students will observe a range of objects, each involving energy. Explain to students that it doesn't matter if they don't yet use scientific terms; they just need to identify what the energy looks like in that situation.

Step 3: **Brainstorm** with the whole class what sorts of words are commonly used when talking about energy. Note: The **Find out more** link in the *Student Digital* has a comprehensive list of energy types.

Step 4: Instruct students (in pairs or trios) to record their observations in their **Notebook** by writing down the label identifying each activity and listing words they think describe the energy forms they see.

Step 5: Move groups around the stations, ensuring all have time at each.

Step 6: Reform the class to share observations. Ask each group in turn for the energy words associated with each activity, drawing on their **Notebook** entries.

Suggested questions:

Key Question - What features told you energy was involved in each activity?

This asks students to make a generalisation; a general principle or idea linked to a range of observations.

NOTE: Typical answers include; things changing, things happening, things moving. The key idea here is change. Energy transformations always involve observable change.



1.2 Energy to burn

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy transfers involve change.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.

Lesson outcomes

At the end of this activity students will be able to:

- share their ideas about what types of energy are visible during a rocket launch
- recognise that various changes indicate energy is involved
- identify that energy can be transformed.

What ideas might your students already have?

- students may be unfamiliar with identifying energy in everyday situations.
- students may associate energy only with living things.
- students may associate energy only with movement.
- students may equate energy only with fuel.

Key vocabulary:

Kinetic energy, gravitational potential energy, chemical energy, energy transfer.

Equipment list

Each CLASS will require:

- *Student Digital*

Each STUDENT will require:

- **Notebook**

Things to consider

Some forms of energy students might mention may overlap, such as movement (bulk movement of the rocket) and vibration. These are obvious examples of kinetic energy, but are different. Accept all different suggestions.

This activity should engage students, so allow their ideas free range. They should feel good at identifying when energy is at work.

Teacher content information

Forms of energy that could be identified include:

- sound



- light
- heat
- movement (kinetic energy)
- falling down (ice falling as it breaks away from the rocket)
- gaining height (escaping gravity; potential energy)
- gaining speed
- vibrations (one astronaut talks about this)
- electricity (there were many cables disconnecting and mention of switches)
- burning chemicals (lots of smoke)
- ice melting.

Lesson plan

Step 1: Ask the whole class what they think energy looks like. How can they tell if they are seeing energy? They may successfully identify situations where change is occurring. They may also identify some accepted forms of energy.

Note suggestions on the board and save for later referral.

Step 2: Tell students they will see a video of an historic rocket launch, Apollo 11 in 1969. Although the video might seem to them very old, it still shows the most powerful engine ever built. This is a good video as it involves obviously large amounts of energy.

Students should work in groups of two to four; whatever is convenient in your classroom.

In their **Notebook** students should write all energy examples. You can show the video twice for students to expand their lists.

Step 3: Run a small competition. Each group, in turn, offers one energy example from the video. Record them on the board, allocating one point for each new **Notebook** example.

Each group must offer a new example. On successive rounds it will be harder to find new examples.

Step 4: Teacher-led discussion. Ask students if they noticed that energy made things happen or produced changes. For example, light and sound where there was none before, movement, gaining height, shaking and chemical reactions.

Help students summarise identified types of energy.

- movement (kinetic energy); could include vibrations
- light
- sound
- heat
- chemical
- gaining height (also called gravitational potential energy)



1.3 Forms of energy

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy transfers involve change.
- Energy looks different in different situations – it can be transferred from one object to another.

ACSI148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- identify energy forms in a photograph using appropriate scientific terms
- clearly label a photograph so that information about energy forms is clearly communicated to another person.

Equipment list

Each PAIR OF STUDENTS will need:

- magazine picture of reasonable size. You can ask students to bring their own magazine, but it is wise to keep a supply handy.
- scissors, glue, pens or pencils for labelling
- butchers' paper.
- **Things to consider**
- you may need to hint if students miss less obvious energies: e.g. unless the picture is totally black, it includes light; is anything producing sound?
- if the picture contains any living thing, it will be using energy.
- if the picture contains plants, they will be absorbing sunlight.
- each student should have their own picture, but could work in small groups to complete the task.
- this activity could be started in class and completed as homework.

This activity could be used as **formative assessment** and repeated later in the unit, or a similar activity could be set as a final test.

Lesson plan

Step 1: Ask students to select a picture. It does not need to be action packed; a quiet scene will be full of energy examples. The picture should be glued in the centre of a piece of butcher paper.

Step 2: Ask students to clearly label every example of an energy phenomenon. The space on the paper around the image should be used. Labels should be large and clear.

Step 3: The images should be displayed around the room so the class can share their images and ideas.



2.1 Throw the ball

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy looks different in different situations – it can be transferred from one object to another.
 - Objects can have mechanical energy because of how they are moving or because of their position.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- apply the concepts of kinetic and gravitational potential energies to explain the throwing of a ball
- explain how the concept of conservation of mechanical energy accounts for the motion of the ball.

What ideas might your students already have?

- Students may not realise the motion slows on the way up and accelerates on the way down
- Students may believe the motion requires the continuous application of a force on the way up, but not on the way down.

Key vocabulary

Kinetic energy, gravitational potential energy.

Equipment list

The CLASS will need:

- a ball (e.g. tennis ball) for throwing, and a good place to view the throw.

Things to consider

If you have a balcony, you could demonstrate throwing the ball from the ground up to balcony height, then watching it fall, compared to simply dropping the ball from the balcony. Qualitatively, the fall should look the same. A catcher might judge it felt the same in both cases.

Some students could observe from the ground, while others observe from the balcony.

This is an opportunity to engage a student with good throwing and catching skills. A video of this is provided in Science by Doing Student Digital and could be observed in class.



Teacher content information

Kinetic energy (KE) is initially provided to the ball while it is still in contact with the throwing hand. Once the ball leaves the hand there is no input of energy. As the ball rises the KE is transformed to gravitational potential energy (GPE). This GPE is effectively stored in the height of the ball. (Strictly it is stored in the greater distance between the ball and the centre of the Earth.)

At the top of the flight the ball stops momentarily (a difficult concept) at which point it has zero KE and maximum GPE.

An important observation is that the flight up and the flight down are symmetrical; each is a reverse of the other. This relates to the conservation of mechanical energy in this system.

Lesson plan

Step 1: Take the class outside and ask the best thrower to launch a ball vertically as high as possible.

Ask the class to observe the specific sequence of events. They could use their mobile phones to video the activity.

Establish that the ball gets an initial push from the thrower, then gets slower as it rises, picking up speed again as it comes down.

Step 2: Direct Instruction - Describe how the thrower gave the ball an initial energy burst by giving it speed. Establish that this is called kinetic energy. The ball's entire energy during the flight was provided in this throwing phase.

As the ball rose it gained height. Establish that we call this gravitational potential energy.

As the ball falls, and loses height, it picks up kinetic energy again.

Together kinetic and gravitational potential energies are referred to as *mechanical energy*.

Provide students with suitable notes outlining these ideas for their **Notebook**. They will apply these ideas in the **Activity 2.2: Energy skate park**.



2.2 Energy skate park

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy looks different in different situations – it can be transferred from one object to another.
- Objects can have mechanical energy because of how they are moving or because of their position.

ACSIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- apply the concepts of kinetic energy and gravitational potential energy to explain the motion of a skater
- explain how the concept of conservation of mechanical energy can be used to account for the motion of a skater.

Key vocabulary

Kinetic energy, gravitational potential energy, simulation, conservation.

Things to consider

Students should observe that adding mass to the skater does not make any difference in the simulation. Some students may wonder whether adding mass doesn't add to gravitational potential and kinetic energy. It does, however, in this situation, they both cancel out: the extra gravitational potential energy is exactly counteracted by the extra kinetic energy at the bottom, due to extra mass.

Students can also run the simulation with or without friction.

Teacher content information

It appears that, without friction, when the skater runs up the other side, he or she keeps going until the starting height is reached.

What happens in the middle example, where there is just one downward slope? How far will he travel before he stops? Galileo, more than 300 years ago, asked himself this question. He experimented rolling balls down slopes. He came to the startling conclusion that the ball would roll forever, unless something stopped it. This idea developed into Newton's first law of physics: the law of inertia.

Lesson plan

Step 1: Have students explore the **Energy skate park** application. Explain it is a simulation, which can be set to make the skateboarder's ride frictionless. This is impossible in real life.

They should first run the simulation with zero friction.



Notebook questions:

- How high up the ramp do you have to start the skater so he or she can get over the middle hump?
- Where is gravitational potential energy the greatest?
- Where is kinetic energy the greatest?
- What happens if you change the skateboarder's mass?

Step 2: Run the simulation again with friction. You can also adjust the mass.

Describe in your **Notebook** how this affects the skater's movement.

Step 3: (Class discussion)

A scientific theory says that, if there is no friction, mechanical energy is conserved in a system.

Explain how this idea can account for what happened in the Energy skate park simulation.

Extension question:

It appears that, without friction, when the skater runs up the other side, he keeps going until he gets to the original height he started from. What happens in the middle example where there is just one downward slope? In this simulation, how far will he travel before he stops?



2.3 Energy in, energy out

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy is the capacity to do work.
- Energy can't be created or destroyed.
- Energy looks different in different situations – it can be transferred from one object to another.
- Objects can have mechanical energy because of how they are moving or because of their position.

ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- identify **effort** and **load** in pulley systems
- explain the concept of **work** as applied to pulley systems
- explain the concept of **mechanical advantage** as applied to pulley systems
- calculate work in and work out in pulley systems and apply the concept of conservation of mechanical energy.

What ideas might your students already have?

- Students might recognise that pulley systems are good for multiplying force but will find it more challenging to apply the concept of work to realise that energy is not also multiplied.

Vocabulary:

Mass, weight, effort, load, mechanical advantage, conservation of mechanical energy.

Things to Consider

The measurements and calculations depend on students having completed **Activity 2.3** so that they can confidently handle forces measured in Newtons.

Teacher content information

Simple machines usually achieve one, or both, of two things. They change the direction of a force and/or multiply force. The pulley system typically changes the direction of a force, so that a downwards force lifts an object. The advantage is body weight can be used as a force to lift objects.

Pulleys can also multiply force. In Experiment 2 a single sheave pulley is used and the LOAD is supported by two ropes. For the load to rise 1 m the effort distance is 2 m, and the effort force is only half the load force.

The multiplication of force is called the mechanical advantage. The mechanical advantage of a pulley system can be calculated either by:

Energy

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▲ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

▲ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

load/effort, or

- count the number of ropes supporting the moving pulley block holding the load.
(Hence the pulley in Experiment 2 has a mechanical advantage of 2.)

This activity provides the most powerful example so far of conservation of mechanical energy.

The work done by the effort equals the work done on the load.

This applies a simple, but powerful, energy concept of work.

$$\text{WORK} = \text{FORCE} \times \text{DISTANCE}$$

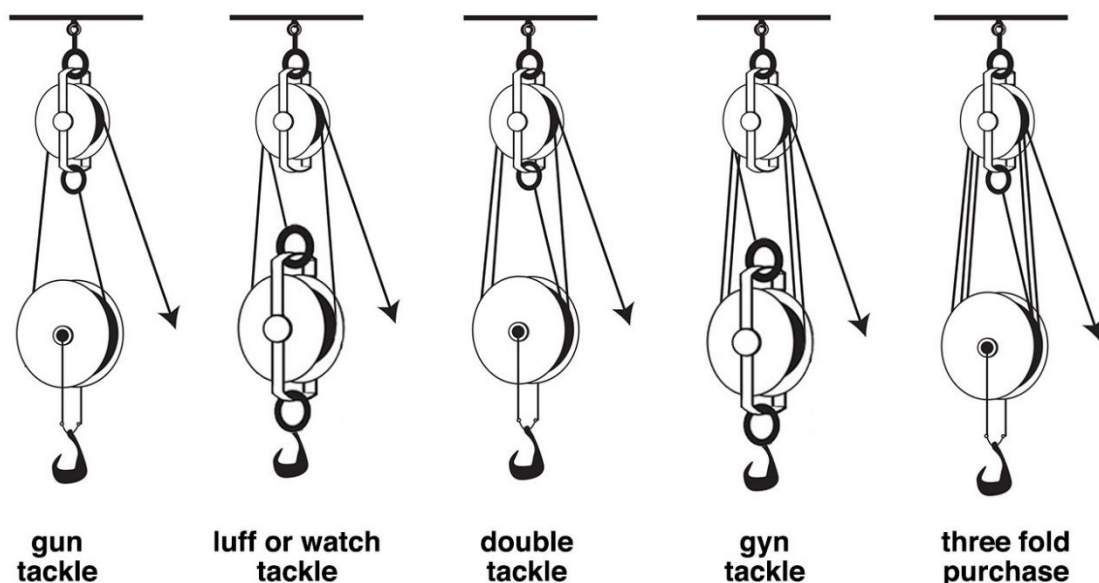
(Work is measured in *Joules*, force in *Newtons* and distance in *metres*.)

Lesson plan

Students should carry out the activities described in **Activity Sheet 2.3**.

They conclude by designing and testing two pulley systems.

The following diagram shows several possible block and tackle arrangements. The term block and tackle is applied to many pulley systems incorporating both fixed and suspended sheaves.



Suggested questions:

On the worksheet fill in the results table.

You will note a column for you to multiply the force (in Newtons) and the distance (in metres). Why is this calculation included?



2.4 Heat from friction

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- The temperature of an object can be raised by doing work on it (e.g. friction).
 - Temperature is related to the movement (kinetic energy) of the particles in a substance.
- AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain that movement energy can be converted to heat through friction
- explain that the transfer of heat energy can be recognized through the rise in temperature of a substance.

What ideas might your students already have?

- Students will probably be comfortable with the idea that friction involves contact between physical objects.
- Students may not recognise that friction is a force, which makes it difficult for them to understand that friction involves doing work on the system.

Equipment list

Each GROUP will require:

- graduated measuring cylinder
- water
- two empty metal soup cans, labels removed
- thermometer (0 - 100 °C)
- at least 1 m of thick string
- oven glove
- ice-block sticks for stirring.



Things to Consider

This experiment involves the transfer of energy from motion (kinetic energy) to heat energy. It also involves measuring heat transfer from string to metal to water by measuring temperature rise. The relationship between heat and temperature will be explored in greater detail in **Part 3**.

Teacher content information

The action of the string effectively does work on the system. This is a direct application of the definition introduced in **Activity 2.3**. In this case it is impossible to measure the work directly, but frictional force and movement are involved, a combination which fulfils the definition of work.

It also introduces the terms *heat* and *temperature* for discussion. The friction between the string and the can produces heat, which is transferred to the water. This transfer is detected as a rise in the temperature of the water.

Lesson plan

What to do

Step 1: Pour 25 mL of water into each can and measure the starting temperatures.

Step 2: Loop about 1m of string around one can. A student wearing the glove holds the can very still. See the *Student Guide* for a diagram.

Step 3: Pull the cord backwards and forwards about 30 times, holding it as tightly as possible.

Step 4: Quickly stir the water and remeasure the temperature of the water in each can.

Step 5: Students should record their results in their **Notebooks**.

Suggested questions:

- Was there a difference in water temperature between the cans?
- Did you put mechanical energy into the string? Did it move?
- Did you put mechanical energy into the can? Did it move?
- Did you apply much force when you moved the string back and forth? What influenced the amount of force you applied?

What happened to the energy you put in?



2.5 Starting fires

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy transfers involve change.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - The temperature of an object can be raised by doing work on it (e.g. friction).
- ACSHE226 Science knowledge can develop through collaboration across the disciplines of science and the contributions of people from a range of cultures.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- apply their understanding to explain that fire required:
 - mechanical energy to cause friction, producing heat up to ignition temperature
 - fuel with a low ignition temperature
 - oxygen
- relate these ideas to how people in the video clips made fire using the rubbing and drill and bow methods.

What ideas might your students already have?

- Students will probably know of some ways of making fire that were used in the past.
- Students may not realise how recently current techniques for starting fires were invented and adopted.
- Students may not distinguish between the concepts of air and oxygen, and may not believe exhaled air contains oxygen.

Things to consider

This activity introduces an indigenous perspective to the central human problem of starting fires. You could briefly discuss how fire and cooking have been integral to survival since modern humans evolved, and probably longer. It is worth noting that when European colonists first settled in Australia their methods for starting fires were different to, but not more efficient than, the methods being used by the local inhabitants.

Teacher content information

All the fire-starting techniques rely on:

- producing high-temperature embers to reach ignition temperature, either directly through friction or indirectly through percussion between a piece of iron and hard stone, such as flint



- a fuel with a low-ignition temperature to start the fire, known as *tinder* or *kindling*
- oxygen, often provided by blowing on the tinder.

Students often don't realise exhaled air contains oxygen. You can link this to mouth-to-mouth resuscitation, which wouldn't work if exhaled air lacked oxygen. It is actually about 16 per cent oxygen, slightly less than the 21 per cent in the atmosphere.

Possible extension activity: nothing is more impressive than seeing someone light a fire with a bow and drill set. Try it: it takes skill, practice and correct equipment. The following video has step-by-step instructions.

<http://www.youtube.com/watch?v=DAs2xnfNidM>

Remember that people usually carried a set of fire sticks, rather than making them new each time. Also, mastering the art takes practice, so don't make your first attempt in class.

It's a challenge, but your class would be very impressed! Remember to fill in an appropriate risk assessment.

Lesson plan

What to do

Step 1: Students watch the videos and record in their **Notebooks** what is done.

Step 2: As a class, students discuss why each step was taken.

Suggested questions:

Specific questions to relate this to **Activity 2.5** include:

- Why are the sticks rubbed together?
- Why is the ember collected?
- Why do they blow on the tinder?

Help the students summarise in their **Notebooks** how these techniques relate to what they learnt in **Activity 2.5**.



2.6 Putting it together

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy transfers involve change.
- Energy looks different in different situations – it can be transferred from one object to another.
- Objects can have mechanical energy because of how they are moving or because of their position.
- The temperature of an object can be raised by doing work on it (e.g. friction).

ACSHE226 Science knowledge can develop through collaboration across the disciplines of science and the contributions of people from a range of cultures.

ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.

Lesson outcomes

At the end of this activity students will be able to:

- apply their understanding of energy forms and mechanical energy transfers to a variety of situations and contexts
- explain a range of energy transformations.

Things to consider

This activity could be completed as a class discussion (to revise ideas) before students attempt the quiz in the **e-Notebook**, which should take 30-40 minutes.

Lesson plan

What to do

Step 1: Discuss *Student Guide Activity 2.6* as a class for about 10 minutes.

Step 2: Each student completes the quiz.



3.1 What is heat?

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy isn't stuff.
- Energy looks different in different situations – it can be transferred from one object to another.
- The temperature of an object can be raised by doing work on it (e.g. friction).
- Temperature is related to the movement (kinetic energy) of the particles in a substance.

ACSHE226 Science knowledge can develop through collaboration across the disciplines of science and the contributions of people from a range of cultures.

Lesson outcomes

At the end of this activity students will be able to:

- describe how Count Rumford's observations provided evidence against the caloric theory of heat.

What ideas might your students already have?

- Students will have a general understanding that heat flows from hot objects to cold. This will be covered in **Activity 3.3**. The common use of the word flow is unfortunate, as it suggests a fluid.

Key vocabulary

Heat

Teacher content information

It is not the most desirable approach to introduce weapons of war as teaching tools, but this particular experiment was a fundamental and famous one, which transformed our understanding of heat and energy.

Cannons originated in ancient China along with the gun powder required to fire them. They were important in warfare for many centuries and modern artillery guns are simply a later modification. Only recently have powered missiles replaced modern artillery.

Cannons were traditionally made by casting in solid metal, usually bronze or iron. The barrel was bored out using a special grinding tool. During grinding the metal became very hot. The heat's origin was explained by the caloric theory. Caloric was believed to be a fluid-like substance that contained the heat. It was released as metal was shaved away from the barrel.

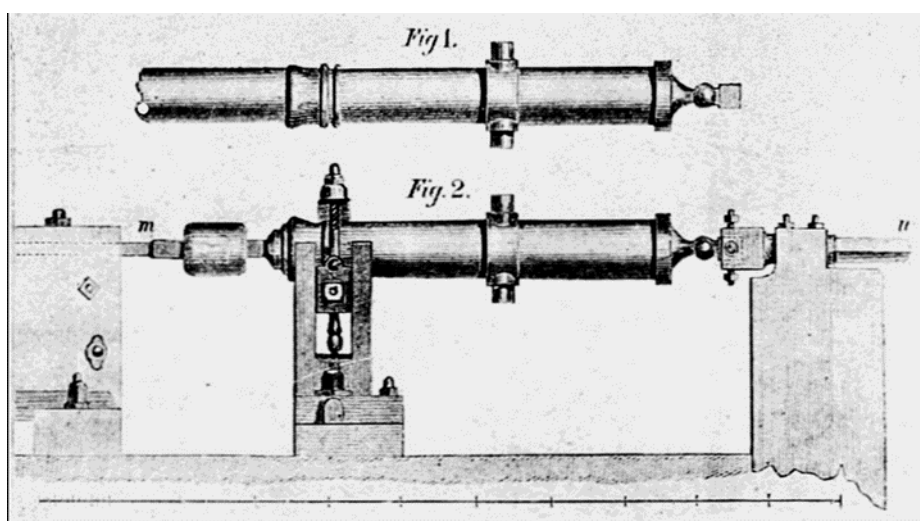
Benjamin Thompson, later known as Count Rumford, undertook a series of experiments into the heating effects of friction. His conclusions were published in an essay, *An Inquiry Concerning the Source of the Heat Which is Excited by Friction*, which were presented to the Royal Society in 1788. [Interesting historical coincidence: this was the year the First Fleet arrived in Sydney Cove].



Count Rumford measured the temperature change as a result of boring a cannon with the apparatus illustrated below.

If caloric was a fluid, it should weigh something, so if it were released, mass should be lost during the experiment. However, the total mass of metal after the experiment equalled the mass of matter before.

By surrounding the end of the cannon with a water filled box, and measuring the temperature rise during the boring process, Rumford proved that a blunt cutting tool produced a large heating effect. He collected the metal cut from the cannon and showed it could not have been the source of the caloric. Further experiments allowed him to rule out the water and the rest of the machinery as the source of the caloric.



He also found more heat was produced using a blunt boring tool, rather than a sharp one. He observed heat generated by friction seemed inexhaustible – which was at odds with the idea that caloric was a substance of limited amount. Apparently, friction could create as much of this substance as desired.

He concluded heat was a form of motion. We now know bodies have heat because of the random kinetic energy of their constituent atoms and molecules.

Lesson plan

What to do:

Step 1: Students should view the animation of Thompson's (Count Rumford) cannon boring experiment in *Student Digital Activity 3.1*. The video has three parts so students can answer questions provided in the *e-Notebook*.



3.2 Keeping your drink warm

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy isn't stuff.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - Heat energy is the energy that flows from hot to cold objects.
 - Temperature is related to the movement (kinetic energy) of the particles in a substance.
 - Energy transfer through different mediums can be explained using wave and particle models.
 - Heat will flow from a hot object to a colder object until they are at the same temperature.
 - Heat can be transferred through conduction, convection or radiation.
- AC SIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.
- AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe mechanisms of heat transfer: convection, conduction and evaporation.
- apply the concepts of heat transfer through convection, conduction and evaporation to the problem of keeping a drink hot.

What ideas might your students already have?

- Students may have difficulty in appreciating the importance of movement within invisible fluids, particularly air.

Key vocabulary

Heat, temperature, conduction, convection.

Equipment list

Each GROUP will require:

- three identical beakers around 100-150 mL.
- thermometer (0 - 100 °C)
- beaker insulating material such as bubble wrap, newspaper, foam, wool
- hot water.



Things to consider

This activity is a solid introduction to convection and conduction. Radiation, as a form of heat transfer, is not introduced until **Activity 3.4**. Although heat is emitted by all surfaces above absolute zero (the phenomenon of black-body radiation) it does not contribute substantially to heat loss in this situation.

Teacher content information

It is surprising how much heat is lost from the surface of a drink. Covering with a thin piece of paper can significantly reduce this effect. Two processes occur at the surface: evaporation absorbs a large amount of energy and the warm, moist air is then carried away by convection currents. Covering the drink disrupts these currents and hence reduces the rate of evaporation at the surface.

Lesson plan

What to do:

Step 1: Students fill a beaker with hot water. Ask them to touch the sides of the beaker and to hold their hands above it.

Can they feel heat through the glass of the beaker? This is called *conduction*.

Can they feel warm air rising from above the beaker? This is called *convection*.

Step 2: Pose the research question: is most heat lost from the drink through conduction from the beaker's sides or through convection from the drink's surface?

Ask each group to devise ways to minimise each effect. These could include simply wrapping the beaker in an insulating material and covering the top with paper to prevent warmed air escaping. Students should carefully keep the volumes equal so heat loss is clearly demonstrated by temperature decline.

Step 3: Students test their different techniques to find which most effectively maintains the temperature of a hot drink. They must clearly understand how to do this. How will they know if they have seen a real effect? Introduce the idea of a suitable experimental control.

Suggested questions:

- How did the heat loss from the two experimental beakers compare with the heat loss from the control beaker?
- What was happening at the surface of the hot drink that would have contributed to the heat loss?

Help students record a concluding statement in their **Notebook**.



3.3 Water is hard to heat

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy looks different in different situations – it can be transferred from one object to another.
- Heat energy is the energy that flows from hot to cold objects.
- If heat is added to an object, its temperature will rise.
- Heat will flow from a hot object to a colder object until they are at the same temperature.
- Heat can be transferred through conduction, convection or radiation.
- It takes a lot of energy to heat or boil water and to melt ice.

AC SIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.

AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- correctly apply the terms *heat* and *temperature*
- understand that not all objects of the same mass and at the same temperature have the same heat content* (refer Teacher Content Information)
- understand that heat is transferred from objects at a higher temperature to objects at a lower temperature.

What ideas might your students already have?

- Students struggle to distinguish between the concepts of heat and temperature.

Key vocabulary

Heat, temperature, thermal equilibrium.

Equipment list

Each GROUP will require:

- foam cups
- hot water and tap water
- 100 mL measuring cylinder
- thermometers (0 - 100 °C)
- 50 g brass metal masses from a slotted mass kit

Energy



▲ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

▲ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

- ice-block sticks for stirrers.

Things to consider

This activity explores the transfer of heat between substances, leading to temperature changes. The key idea explored is that objects in contact will come to the same temperature, regardless of how hot or cold they feel.

Teacher content information

It is more correct to refer to the internal energy or thermal energy of a system that has absorbed heat energy, but this terminology can be confusing at this stage, so we refer to heat content. What is important is that if a system absorbs a certain amount of heat and its temperature increases, and no other changes occur, such as a change in volume, then all this heat will be available in the reverse reaction.

This activity explores the transfer of heat between substances, leading to temperature changes. The key idea being explored is that objects in contact will come to the same temperature, regardless how hot or cold they feel. Two objects at thermal equilibrium will be at the same temperature.

Water has an enormous capacity to absorb and deliver heat without large changes in its temperature. This property is referred to as specific heat (or thermal) capacity. The specific heat capacity of brass is much lower than that of water (around 0.4 rather than 4.2 J/g/ °C). The value for water is higher than most other known substances. In the second experiment, the brass weights will have a relatively small effect on the temperature of the hot water. Theoretically it would take nearly a kilogram of brass to equalise the temperature of 100 mL of water.

This property of water leads to many of its important characteristics, such as the difficulty of heating water, large electricity bills associated with frequent and long showers, moderate climate on islands, Earth's relatively stable climate, and the stable temperatures of large animals and the use of water in extinguishing fires.

Lesson plan

What to do:

Experiment 1

Step 1: Add 100 mL of hot water to one cup and measure its temperature.

Add 100 mL of cold tap water to another cup and measure its temperature.

If you mix these two amounts of water together in one cup, what temperature will the mixture will be? Why?

Step 2: Mix the two amounts of water and test your hypothesis. Stir quickly with a stick and measure the new temperature.

What did you find? Perhaps the results were unsurprising?

Experiment 2

Now mix two different substances, one water and a metal, brass.



What will happen if you add 100 g of cold brass to 100 mL of hot water? As in the first experiment, they have the same mass, as 100 mL of water has a mass of 100 g, so you will still be mixing the same mass amounts.

Predict a result in your **Notebook**.

Step 1: Bring the temperature of the brass masses (100 g, usually two 50 g masses) to equal that of cold water by immersing them in tap water, and measure the temperature.

Step 2: Add 100 mL of hot water to a cup and measure its temperature.

Step 3: Use tongs to quickly transfer the cold masses to the hot water. Stir with a stick and measure the new temperature.

Suggested questions:

- In **Experiment 2** did the hot water cool? Where did the heat go?
- Was there a difference in the capacity of the cold water and the cold brass to cool down the hot water?

Extension

Students could design an experiment to approximate the difference in heat capacities of brass and water. They could use a larger volume of water and test what mass of brass was required to bring its temperature to half way between the original brass and water temperatures.

A comparison with water is a standard approach to determining specific heats. Without calculations, only an approximate value will be reached, but since the specific heat of water is more than ten times that of brass, the effect should be obvious.

Help the students record a concluding statement in their **Notebook**.



▲ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

▲ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

3.4 Feeling hot and cold

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Heat energy is the energy that flows from hot to cold objects.
 - Temperature is a measure of how hot or cold an object is.
 - If heat is added to an object, its temperature will rise.
 - Temperature is related to the movement (kinetic energy) of the particles in a substance.
 - Energy transfer through different mediums can be explained using wave and particle models.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- AC SIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.
- AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- explain how the conduction of heat relates to whether surfaces feel hot or cold
- understand objects in contact in the same environment will come to the same temperature.

What ideas might your students already have?

- Most students will have the misconception that metal surfaces feel cold because they are at lower temperatures.

Key vocabulary

Heat, temperature, conduction, senses.

Equipment list

Each GROUP will require:

- metal plate (e.g. retort stand's base, unpainted)
- plastic of similar size (e.g. small plastic chopping board)
- ice cubes
- egg rings

Things to consider

See the video in *Student Digital Activity 3.4* for the best way to conduct this experiment. Students may be inspired, and able, to conduct this experiment at home on their own subjects.



The metal and plastic plates, and the rings, used in the video are available from Arbor Scientific [<http://www.arborsci.com/ice-melting-blocks-thermal-conductivity>].

Teacher content information

Our sense of touch, in detecting hot and cold, depends on our skin absorbing or losing heat. This may occur when we touch a surface at a very different temperature or one at only a slightly different temperature, but with high thermal conduction. Metals are good thermal conductors (as well as good electrical conductors).

The space shuttle heat tile is non-metallic and conducts heat poorly. Little heat is lost to the surrounding air (or to the hand) through conduction. It is black and therefore emits radiant energy very well. Most of its heat loss occurs through radiation, mostly in the infra-red range.

Lesson plan

What to use:

Each GROUP will require:

- large container or sink half filled with water
- large ball (hollow)
- spring balance or force measurer
- two masses tied to a string (one that would sink and one that would float).

What to do:

Step 1: Have students view the video in *Student Digital Activity 3.4*.

Step 2: Provide appropriate materials so students can replicate Dr Muller's experiment in the classroom. Everyday objects are best so students can repeat the experiment at home. This could be a whole class demonstration.

Suggested question/s:

Explain, using the term thermal conduction, why the person pictured in **Activity 3.4** can hold the red-hot space shuttle heat tile in his bare fingers.



3.5 Heating water with light

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy transfers involve change.
 - Energy isn't stuff.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - Energy transfer through different mediums can be explained using wave and particle models.
 - Heat can be transferred through conduction, convection or radiation.
- AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- demonstrate heating water with light
- provide evidence that dark surfaces absorb more light energy than light-coloured surfaces.

What ideas might your students already have?

- Students may be familiar with the effect of wearing a white or black shirt on a sunny day.

Key vocabulary

Heat, temperature, light, radiation.

Equipment list

Each GROUP will require:

- black and white spray paint
- three empty aluminium drink cans: one spray painted black, one painted white, one wrapped in foil. You could include cans painted with shiny and matt black paint. Cans should be painted beforehand.
- thermometers (0 - 100 °C)
- tap water
- sunny day.



Things to consider

A class set of control cans will also be needed.

You will need a sunny outdoor spot where the class can settle down for half a lesson as the cans heat up.

The final step includes combining all the groups' results on the board to produce a class result. This lets you introduce the idea of replications in an experiment to improve accuracy and reliability. Several groups provide replicates, rather than each group replicating the experiment several times.

Teacher content information

By *solar energy* we mainly mean light energy, mixed with some infra-red radiation. This is the radiation absorbed by surfaces producing heat. As in other experiments in this unit, heating water is a standard way of detecting heat transfers.

The black cans will absorb most of the light that falls on them, while the white and alfoil-covered cans will reflect most light. There should be no difference between the controls if they are in complete shade.

Give some consideration to where to place the cans in the sun. Ensure they are not placed on a hot surface where they will absorb heat from conduction.

Lesson plan

Step 1: Add 250 mL cold tap water to each can and measure the temperature. It is preferable if all cans are close to the same temperature.

Step 2: Place the cans side by side in direct sunlight so all, if possible, receive the same light exposure.

The class control set, including one can of each colour, should be placed in complete shade.

Step 3: Monitor the temperature of the cans every five minutes for 30-40 minutes or until a significant temperature increase has occurred in at least one.

Step 4: Determine the temperature increase in each can. Combine the results of each group on the board and ask the class to calculate the average for each can type.

Students should report the results in their **Notebook**, including a diagram of the setup.



3.6 Solar energy

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy transfers involve change.
- Energy isn't stuff.
- Energy looks different in different situations – it can be transferred from one object to another.
- Energy transfer through different mediums can be explained using wave and particle models.
- Heat can be transferred through conduction, convection or radiation.
- Most nations are exploring ways of providing renewable energy to cities.

AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

AC SIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.

AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- apply an understanding of heat and light to design a solar collector to successfully heat water
- evaluate the effectiveness of a solar heater by taking appropriate measurements.

Key vocabulary

Heat, temperature, light, radiation, reflection.

Equipment list

Each GROUP will require:

- cardboard box (large enough to house your collector)
- thermometer (0 - 100 °C)
- cling wrap
- alfoil
- plastic PET bottle, around 1.25 L
- sunny day



- students may request other useful equipment.

Each STUDENT will require:

- **Activity Sheet 3.6 Solar energy**

Things to consider

This activity is an opportunity for students to apply, and for you to evaluate, their understanding so far in the unit. It could be used as a summative assessment item. Students will be asked to write a report describing their design, using appropriate scientific language to explain why it was or wasn't effective.

A basic bottle-in-a-box design can be used, or the more complex circulating system shown at the following website. However, such systems are difficult to seal and may leak badly.

<http://www.behance.net/gallery/SOLAR-WATER-HEATER-EXPERIMENT/2753215>

A simple design is shown in this image.



Lesson plan

Distribute **Activity sheet 3.6**. It provides students with instructions and a marking scheme.

Suggested question:

Students should prepare a report describing their investigation and evaluating their design.



3.7 What is sunlight?

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy isn't stuff.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - Energy transfer through different mediums can be explained using wave and particle models.
- ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain that sunlight is a mixture of visible light and invisible ultraviolet and infra-red light.
- describe the visible spectrum
- understand that light is a way energy can be transferred through the vacuum of space.

What ideas might your students already have?

- Students will know light comes from the sun, but might not know that a significant amount is infra-red light or heat rays.

Key vocabulary

Heat, light, radiation, spectrum.

Equipment list

Each GROUP will require:

- glass prism and a ray light source, such as from a standard light-box kit.

Things to consider

This activity allows students to visualise light and its spectrum. It will link to previous activities where they used sunlight to heat water. Students will understand the sun emits light, the only energy form that reaches the Earth's surface.

Teacher content information

Light has a spectrum, which is centred on visible light. Ultraviolet (UV) and infra-red are also forms of light, but invisible to human eyes.



Lesson plan

Step 1: Give access to *Student Digital Activity 3.7*.

Step 2: Give students a prism and suitable light source to produce their own spectrum.

Suggested questions:

Students should complete the **Notebook** questions.



⚠ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

⚠ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

3.8 Cyclones

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy transfers involve change.
- Energy looks different in different situations – it can be transferred from one object to another.
- Heat energy is the energy that flows from hot to cold objects.
- Temperature is a measure of how hot or cold an object is.
- Temperature is related to the movement (kinetic energy) of the particles in a substance.
- Energy transfer through different mediums can be explained using wave and particle models.
- Heat will flow from a hot object to a colder object until they are at the same temperature.
- Heat can be transferred through conduction, convection or radiation.
- It takes a lot of energy to heat or boil water and to melt ice.

Lesson outcomes

At the end of this activity students will be able to:

- describe the overall structure of a cyclone
- explain how heat is involved in its formation
- explain how the evaporation and condensation of water contribute to these heating effects.

Key vocabulary

Heat, temperature, cyclone, hurricane, evaporation, condensation.

Things to consider

This activity introduces the central importance of heat in the Earth's weather and climate systems.

Teacher content information

The evaporation of water to form water vapour absorbs large amounts of energy. This is the basis of evaporative cooling. Conversely, as water condenses back to the liquid state, it releases this heat energy in large quantities.

The most important idea is that cyclones, and much of our weather, are driven by heat. Heat exchange between the atmosphere and the oceans is the principal driver of many weather and climate systems. Oceans warm as they absorb sunlight, causing water to evaporate. As warm, humid air rises, it expands and cools. Water vapour condenses in the atmosphere and releases large amounts of heat, which drives many weather systems.

Lesson plan

Provide access to *Student Digital Activity 3.8*.

Suggested questions:

Students should answer the **Notebook** questions.



4.1 Making a globe light up

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy transfers involve change.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - Electricity is a major way of transferring energy to our cities and homes.
- ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe the requirements for lighting an electric globe
- understand every electric circuit requires a source of electrical energy and a complete circuit.

What ideas might your students already have?

- Students may not understand that electrical devices require a complete circuit to work. This relates to the belief that electricity is delivered to a device, such as a bulb, and used up.

Key vocabulary

Electricity, voltage, positive and negative terminals, circuit.

Each GROUP will require:

- small light bulb without bulb holder (2 V)
- plastic coated wires without the ends stripped
- 1.5 V battery
- pair of wire strippers for removing the plastic from the ends.

Things to consider

This activity introduces electrical circuits. The minimal equipment ensures students can clearly see what constitutes the basic requirements of an electric circuit. This activity avoids using the usual power supply and bulbs in holders to remove black-box effects, so students see exactly what is required. Students will go on to use the more usual electrical equipment in subsequent activities.



Lesson plan

What to use:

Each GROUP will require:

- spring balance (up to 1 kg or 10 N)
- string
- 2 pulleys to attach to end of bench
- 2 x 500 g masses with mass hanger

What to do:

Step 1: Give each group the basic equipment.

Step 2: Instruct the class to make the light bulb glow.

Step 3: When most groups have been successful, ask the whole class what was required.

Suggested question/s:

What were the minimum requirements for lighting your globe?

Wires had to be connected to the battery's *positive* and *negative* terminals. Explain these are indicated by plus and minus signs.

The wires had to be connected to two different parts of the globe. Electricity had to be able to flow in and out of the globe. *A complete circuit was required.*

Students should answer the **Notebook** questions.



4.2 Showing electric circuits

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Electricity is a major way of transferring energy to our cities and homes.

AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- interpret circuit symbols.

Key vocabulary

Circuit, diagram, symbols

Equipment list

Each **GROUP** will require:

- large sheet of paper
- marker pens
- Blu Tack to display poster.

Teacher content information

This activity allows students to develop ownership and understanding of how we communicate with each other. They explore ideas and reach a common understanding. You may increase the number of symbols used (battery, bulb, wires, switch, buzzer, ammeter, electric motor).

However, symbols are not universally agreed and vary for different applications.

Lesson plan

Step 1: Form groups of three or four and ask them to recall the last activity (lighting the globe) and to suggest a set of symbols to represent the equipment used. You may add more, such as a motor or meter. Students brainstorm ideas for symbols.

Step 2: Students write ideas on the paper (using their symbols) to represent the circuit and then perform a **Gallery Walk**.

Step 3: Encourage students to share their ideas from the walk. What were common features? Why?

Step 4: Compare students' ideas to the conventional table of symbols (*Student Guide Activity 4.3*). Discuss as a class why these symbols are standardised.

Suggested questions:

- Why did you use ...?
- Is that the quickest way?
- Would you use ... regularly?

Follow up:

- bingo activity to refresh memory of the symbols



4.3 Heating with electricity

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy is the capacity to do work.
- Energy transfers involve change.
- Energy looks different in different situations – it can be transferred from one object to another.
- Electricity is a major way of transferring energy to our cities and homes.
- Heat energy is the energy that flows from hot to cold objects.
- Temperature is a measure of how hot or cold an object is.
- If heat is added to an object, its temperature will rise.
- Energy transfer through different mediums can be explained using wave and particle models.
- Heat will flow from a hot object to a colder object until they are at the same temperature.
- Heat can be transferred through conduction, convection or radiation.
- It takes a lot of energy to heat or boil water and to melt ice.

ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- explain how a circuit can be built to heat water
- understand most electric circuits transform some electrical energy to heat
- calculate the quantity of heat transformed in a circuit.

Equipment list

Each **GROUP** will require:

- standard 12 V power supply
- electrical leads with alligator clips
- voltmeter (0 - 12 V)
- ammeter (0 - 5 A)
- 20 cm high-resistance wire (Eureka or Nichrome)
- large test tube for winding the heating coil
- tap water
- measuring cylinder
- foam cups
- thermometer (0 – 100 °C).



Things to consider

The success of this experiment depends on the efficiency with which the heat energy generated by the wire is transferred to the water. It is important that:

- as much of the wire as possible is immersed in the water as possible. Make sure that the whole coil is below the water surface. There will need to be a small stretch of wire out of the water for the electrical connections.
- the wire is not in contact with the Styrofoam, which could melt as the wire heats up.
- the loss of heat is minimised. This can be achieved by covering the cup with a piece of card or even paper.

To help prevent the analogue ammeter from burning out, instruct the students to use the 5A port before using the other ports. It is a good idea to put the bulb from a Pasteur pipette over those other ports to prevent accidental use.

Teacher content information

Students are studying the electrical equivalence of heat. The relationship for heating water is:

Power = $V \times I$ (measured in Watts).

The high-resistance wires usually used are either Eureka (a nickel/copper alloy) or Nichrome (a nickel/chromium alloy).

Lesson plan

Step 1: Students examine the directions for setting up the circuit, as shown in the *Student Guide Activity 4.2*. Information about each component and its symbol is provided.

Step 2: Different groups should be assigned different voltages, ranging from 2 V to 12 V.

Step 3: Each group should carefully measure the temperature of 150 mL of water before and after running the experiment for 15 minutes.

Step 4: Each group should add their results to the table on the board (see below).

Table of results

Voltage	Current	$V \times I$ (Power in Watts)	Temp increase of 150 mL of water in 15 minutes.

Step 5: Students should graph power against the temperature increase of 150 mL water in 15 minutes.

Suggested questions:

- Did the temperature increase of the water correspond to the power of your heater?
- What do you think would have happened to the temperature increase of the water if you had doubled the time of heating?



4.4 Build a motor

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy is the capacity to do work.
 - Energy transfers involve change.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - Electricity is a major way of transferring energy to our cities and homes.
- ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- build a model motor and describe its major components
- describe how electrical energy can be transformed into kinetic energy.

Key vocabulary

Electricity, motor, coil, magnet.

Equipment list

Each **GROUP** will require:

- two jumbo-sized paper clips
- 1.5 V battery
- rubber bands
- 2 m of fine, insulated, electrical wire
- super magnet (neodymium): can be purchased from Scientrific for about \$14 .
- plasticine or Blu Tack
- large test tube for producing the coil.

Things to consider

This activity requires students to closely follow the instructions in *Student Guide Activity 4.3*. Once they have produced a working motor they may be able to modify and improve it.

Take care with the neodymium super magnets. They have extremely strong magnetic forces and it is possible for two magnets to be attracted with such force that they can cause injuries such as pinching skin and causing blood blisters. They should also be kept away from electronic devices. Although they have strong magnetic force they are brittle and can break or chip easily.



Teacher content information

All electric motors run on the principle that a wire carrying an electric current will feel a force if it is in a magnetic field. This is called the motor effect. This effect is quite weak, however coiling the wire multiple times will aggregate the individual forces. The wire must be insulated before it is coiled. All electric motors must have, at least, a coil of insulated wire and a magnet. The magnet can be an electromagnet.

This model is a direct current (DC) motor. Shaving the ends of the wire produces the commutator, which switches the current on once each turn.

Most real motors are alternate current (AC) motors, running directly from the AC power supply. They simply require a sliding contact rather than a commutator.



Lesson plan

- Step 1:** Carefully wind the wire around the test tube about 15 times to produce your coil. It should be finely balanced to turn easily, but doesn't slip to one side.
- Step 2:** With scissors, carefully shave the insulated coating from the same side of each protruding wire.
- Step 3:** Shape each paper clip into a cradle as shown.
- Step 4:** With the rubber band, attach both paper clips to the battery terminals.
- Step 5:** Attach the magnet to the battery's steel jacket.
- Step 6:** Secure the apparatus to the table using plasticine.
- Step 7:** Place the coil in the cradle and start it with a nudge.



Suggested questions:

- Did your motor require a complete circuit?
- What energy transfer is your motor achieving?
- Did you know the electric motor was one of the most important inventions of all time?
They are found everywhere.
- How many motors does your family own? This includes every device (large or small) where electricity makes something move. You should be able to find several.



⚠ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

⚠ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

4.5 Build a generator

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy is the capacity to do work.
 - Energy transfers involve change.
 - Energy isn't stuff.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - Electricity is a major way of transferring energy to our cities and homes.
- ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- ACSHE 135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.
- AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- build a model generator and describe its major components.

Key vocabulary

Generator, coil, magnet.

Equipment list

Each GROUP will require:

- solenoid (each group could also produce their own coil with about 1-2 m of insulated wire)
- strong-bar magnet
- electric leads
- galvanometer or micro ammeter.

Things to consider

This activity requires students to follow the instructions provided in the *Student Guide*. Students will see a small *generator effect* using a micro ammeter or galvanometer.

You can give students a more practical example using a bicycle dynamo. The problem with real, working generators, like motors, is that you can't see the inner workings.



Teacher content information

The generator effect means any magnet moving near a conductor will produce a voltage. This is the basis for all large-scale electricity production (excluding solar cells). Every large power station uses turbines to create electricity. Various fuels, including coal, gas, or nuclear reactions are used to heat fluids, usually water, to produce steam to drive turbines. The Snowy Mountains Hydro-Electric Scheme uses water rushing down pipes (transferring gravitational potential energy to kinetic energy) to drive turbines.

Lesson plan

Step 1: Connect the solenoid and the meter in a complete circuit.

Step 2: Quickly pass the bar magnet in and out of the coil. Watch the meter as you do.

Step 3: Hold the magnet in a stationary position inside the coil. Does the needle move now?

Suggested questions:

- Did the meter needle move?
- Was the effect very large?
- What sort of energy is being transferred to electrical energy?

General Idea: electricity is always produced if a magnet moves near a conductor.



4.6 The lemon battery

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy is the capacity to do work.
- Energy transfers involve change.
- Energy looks different in different situations – it can be transferred from one object to another.

ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- construct a battery made of lemons and copper and zinc electrodes.
- explain that the energy to power the battery ultimately comes from the chemical energy in the metal electrodes.

What ideas might your students already have?

- A very common misconception is that this battery is powered by the lemon. In fact the chemical energy comes from the pure metals. A large amount of chemical or electrical energy was put in to produce the pure metals from their ores.

Key vocabulary

Electrode, battery, LED (light emitting diode).

Equipment list

Each **GROUP** will require:

- 4-6 lemons
- 4-6 pieces of copper and zinc, approximately 1 cm x 3 cm
- LED
- electrical leads with alligator clips
- tray for lemons.

Things to consider

A typical LED requires around 2.2 V to light. A copper/zinc electric cell (in any appropriate electrolyte) can produce up to around 0.8 V). Hence at least three cells need to be connected in series to produce an appropriate overall voltage. We have found four in series is sufficient.



Connecting more of these in parallel should produce more current, but uses a lot of lemons.

Unlike light bulbs, LEDs have a particular polarity. Most have the negative electrode indicated by having the plastic case shaved off to produce a flat edge just above the negative electrode. If connected the wrong way, it won't light.

Teacher content information

In this electric battery, the lemon is simply serving as the electrolyte. The energy to drive the system essentially comes from the zinc, which is oxidized, releasing electrons. Zinc dissolves into the juice forming a zinc solution. Electrons travel through the circuit to the copper. Mostly hydrogen is released at the copper. The hydrogen ions come from the acids in the lemon.

Lesson plan

Step 1: Roll each lemon firmly on the table to release juice. You may have seen Jamie Oliver do this on a cooking show.

Step 2: Cut two slits in each lemon.

Step 3: Push one piece of copper and one piece of zinc into each lemon.

Step 4: Use the electric leads to connect the lemons in a circuit. You should connect the lemons one after the other, so that copper connects to zinc on the next lemon, and so on.

Step 5: Complete the circuit by connecting the LED.

NOTE: the pin under the flat side of the LED should connect to a zinc electrode.

Step 6: Experiment with different numbers of lemons in your circuit to find the optimum number.

Discussion

- You could try this experiment with other fruit or vegetables, or even with salty water.
- You could also try different combinations of metals.
- You could also use a voltmeter to check how much voltage you got from your battery.
- What energy conversion was happening in your lemon battery?
- Sketch your lemon battery in your **Notebook**.
- Based on the information provided in the *Student Guide* **Activity 4.6** summarise where the energy to run your lemon battery came from.



5.1 Food as fuel

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy is the capacity to do work.
- Energy transfers involve change.
- Energy looks different in different situations – it can be transferred from one object to another.
- Most of the food we eat is to provide energy.

AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- calculate from the temperature rise of a water sample the quantity of heat energy a burning nut releases.

What ideas might your students already have?

- Students will find it difficult to relate the obvious release of heat from the burning nut to the extraction of energy that occurs in the body through the processes of digestion and respiration. They will think that digestion is where the energy is released.

Key vocabulary

Heat, specific heat, temperature.

TEACHER DEMONSTRATION

Equipment list

Each CLASS will require:

- pecan nut (or alternative like almond, macadamia)
- thermometer (0 - 100°C)
- paper clip to hold nut
- measuring cylinder (50 mL)
- boiling tube
- Bunsen burner
- retort stand, boss head and clamp to hold boiling tube
- heat-resistant mat
- safety glasses.



Things to consider

This experiment produces smoke and should be conducted in a fume hood. This experiment presents risks involving allergy and anaphylaxis. Alternative foods may be substituted for nuts including Tiny Teddy biscuits. Always read the labels as these may include traces of nuts. Teacher discretion is required.

To minimise risk, it is recommended that the experiment is conducted by the teacher and filmed for students to view. Alternatively, there are a range of You Tube video clips you might like to use.

Teacher content information

This is the third activity that uses the heating of water to measure the energy content of something. Heating water, and using the specific heat of water, has been a standard theoretical and practical approach to comparing different energy forms.

Note: When we eat a nut the body doesn't literally burn the nut, but it does get access to all this energy.

First the nut is broken down by digestion into simpler chemicals, such as glucose, which are transported around the body. Very little energy is released at this stage. The real action happens inside every cell, where a complex series of at least 35 chemical reactions take place to gradually break down these products into CO₂ and water, releasing all the energy. This energy doesn't end up as heat directly, but gets transferred to yet more chemicals, particularly ATP.

Lesson plan

What to do:

Step 1: Set up the apparatus in a fume hood, as shown in the diagram in the *Student Guide Activity 5.1*.

Step 2: Add 30 mL of cold tap water to the boiling tube. Measure the starting temperature.

Step 3: Use the burner to set fire to the nut. It may take a few seconds before the nut will burn on its own.

Step 4: Wait until the nut has completely burned away. (It may be necessary to re-light the nut at some stage if it goes out.)

Step 5: Measure the final temperature of the water.

Discussion:

It takes 4.2 Joules of energy to heat 1 mL of water by 1°C. This is called its *specific heat*.

Calculation of energy released.

(The energy output of the nut) = (Volume of water in mL) x (Temperature rise of the water) x 4.2

The answer will be in Joules. You can convert this to kilojoules (kJ) by dividing by 1000.



5.2 Energy in food

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy is the capacity to do work.
- Energy transfers involve change.
- Energy looks different in different situations – it can be transferred from one object to another.
- Most of the food we eat is to provide energy.

Lesson outcomes

At the end of this activity students will be able to:

- access information about the energy content of foods from the nutritional information provided
- compare the energy contents of different foods, accounting for their different weights
- relate this information to average daily requirements
- use the *Nutrition Panel Calculator* to produce *Nutritional Information Panels*.

Key vocabulary

Kilojoules, carbohydrates, sugars, fats.

Equipment list

Each **GROUP** will require:

- a range of packaged foods of different types
- *Student Digital*.

Optional Activity

Each **CLASS** will require:

- ingredients for cooking pikelets
- electric fry pan.

Things to consider

Students would respond well to cooking pikelets. This is most easily done in an electric fry pan. You will need a non-lab class room to do this in. The ingredients are easily mixed and cooked.

Lesson plan

Step 1: Draw the table shown in the *Student Guide* in your **Notebook**.

Step 2: For each food type refer to the Nutrition Information Panel to work out how much energy is in each. This will usually be given for standard 100 g serves.

Step 3: Determine which types of food molecules contribute most to this energy content. Refer to the food molecules described on this page.

Energy

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▲ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

▲ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

Step 4: Refer to *Student Digital Activity 5.2* to construct your own Nutrition Information Panels.

Discussion: Which food molecule contributes the most energy to foods?



⚠ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

⚠ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

5.3 Who eats what?

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy transfers involve change.
- Energy looks different in different situations – it can be transferred from one object to another.
- Most of the food we eat is to provide energy.
- Food chains and food webs describe how energy passes through living things.

AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- construct and interpret food chain and food web diagrams
- understand that food chains and food webs show how energy is transferred through ecosystems.

What ideas might your students already have?

- Students may be surprised that the Antarctic Ocean region has so much life. They will tend to link warm climates with abundant life, which tends to be true on land but not in the oceans.

Key vocabulary

Food chain, food web.

Things to consider

This short activity is designed to get across one idea, that food chains and food webs describe the transfer of energy through biological systems. Students will have learnt about various cycles, such as the carbon cycle and the water cycle. Food chains describe the one way transfer of energy, originally from sunlight and ultimately ending up as waste heat.

Teacher content information

The most important idea in this unit follows on from earlier activities that emphasised that most food consumption is as an energy source. In fact the transfer of energy through food chains is quite inefficient (around 10% in most cases). This means that at least 10 times as much biomass is required as you go down from one trophic level to the next.

Lesson plan

Students complete the questions posed in the *Student Guide*, introducing them to the link between food chains and food webs.



5.4 Rates of photosynthesis

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy transfers involve change.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - Energy transfer through different mediums can be explained using wave and particle models.
 - Food chains and food webs describe how energy passes through living things.
- AC SIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.
- AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe factors that affect the rate of photosynthesis
- explain the overall photosynthesis reaction and word equation
- design a simple experiment to investigate a factor that affects the rate of photosynthesis.

Key vocabulary

Photosynthesis.

Equipment list

Each GROUP will require:

- 250 mL beaker
- filter funnel
- fresh elodea (usually obtainable from pet shops that sell aquarium fish)
- thermometer



- ice
- light source
- 100 mL measuring cylinder.

Things to consider

Students are provided with an experimental setup in the photo on page 51 of the Student Guide. They might need some scaffolding when designing their investigation. In particular, the design needs to be restricted to asking a question related to one factor that can be easily manipulated (probably temperature or light).

Note that plants like elodea respond better to grow-lights than normal lamps, because grow-lights have more blue light in their spectrum.

Note: Elodea is a noxious weed in some places but an alternative pond weed can be used.

Lesson plan

Step 1: Firstly, project page 50 of the **Student Guide** onto the class screen and recap with students the process of photosynthesis. The sugars produced in plants' leaves provide the energy for all cellular activities in the plant. These sugars also form more complex molecules like starch and cellulose and are the basis for all food chains.

Step 2: Complete the digital activity to view oxygen bubbles given off by aquatic plants.

Step 3: Ask each student group to brainstorm all the factors that could influence the rate of photosynthesis and note these factors in their **Notebooks**.

Step 4: Ask each group to design an investigation of one of the factors. Students should think of a question to answer in this investigation (aim). They may also provide a hypothesis.

Students briefly describe their investigation in their **Notebooks**, including a sketch of the experiment.

Step 5: Students begin the experiment and measure the oxygen collected in the cylinder as they test their factor.



6.1 Where does our energy come from?

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy is the capacity to do work.
 - Energy transfers involve change.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - Electricity is a major way of transferring energy to our cities and homes.
 - Most of the energy required for cities and agriculture currently comes from non-renewable chemical sources. This has environmental effects.
 - Most nations are exploring ways of providing renewable energy to cities.
- ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- ACSHE 135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.

Lesson outcomes

At the end of this activity students will be able to:

- describe the processes that deliver electrical energy to cities and towns
- explain the energy transformations that take place in the production of electricity in power stations
- describe some aspects of electricity production and transmission in their own region.

What ideas might your students already have?

- Students may be unaware of many aspects of how electrical power is delivered to their homes.

Key vocabulary

Electricity, voltage, transmission

Lesson plan

Students complete **Activity 6.1**. This involves interactive on-line activities as well as questions to be answered in their **Notebook**.

Discussion:

Students answer questions in their **Notebook**.



6.2 Energy systems

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy transfers involve change.
- Energy can't be created or destroyed.
- Energy looks different in different situations – it can be transferred from one object to another.
- Energy is everywhere, but not all of it is useful.

ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain the use of Sankey diagrams to illustrate the energy transfer through complex systems
- apply their understanding of a Sankey diagram to propose areas where energy could be saved in a complex system such as the family home.

Key vocabulary

Sankey diagram

Things to consider

Sankey diagrams develop mathematical skills by representing numerical quantities.

Teacher content information

Sankey diagrams summarise all the energy transfers underway in a process. The thickness of each line is proportional to the amount of energy involved. Sankey diagrams are named after Irish Captain *Matthew Henry Phineas Riall Sankey*, who used this type of diagram in 1898 in a publication on the *energy efficiency* of a steam engine.

This Sankey diagram for an electric lamp shows that most of the electrical energy is transferred as heat rather than light.

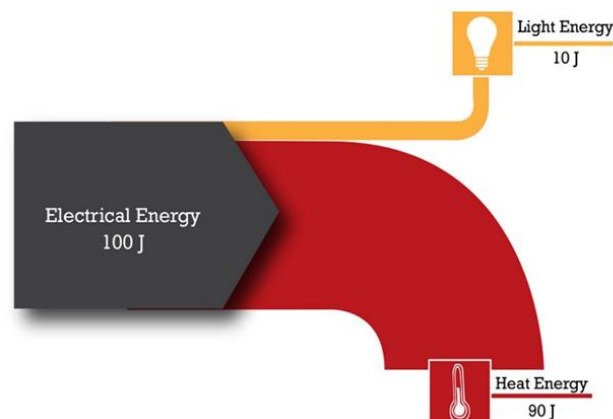


Figure: Sankey diagram for a filament lamp



Energy can be transferred usefully, stored or dissipated. It cannot be created or destroyed. Notice that 100 J of electrical energy is supplied to the lamp. Of this, 10 J is transferred to the surroundings as light energy. The remainder, 90 J (100 J – 10 J) is transferred to the surroundings as heat energy.

The energy transfer to light energy is the useful transfer. The rest is wasted: it is eventually transferred to the surroundings, making them warmer. This wasted energy eventually becomes so spread out that it becomes less useful.

Lesson plan

Step 1: Students should complete *Student Digital Activity 6.2*.

Step 2: Students should be asked to report and explain their reasoning for the final **Notebook** question.

Discussion:

Students will answer questions in their **Notebook**.

The whole class should discuss their answer and reasoning for the **Notebook** question;

Based on this diagram where do you think you could save the most energy overall?



6.3 Conserving energy

Curriculum links

ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.

KEY IDEAS

- Energy can't be created or destroyed.
- Energy looks different in different situations – it can be transferred from one object to another.
- Heat energy is the energy that flows from hot to cold objects.
- Heat will flow from a hot object to a colder object until they are at the same temperature.
- Heat can be transferred through conduction, convection or radiation.
- Energy is everywhere, but not all of it is useful.

Lesson outcomes

At the end of this activity students will be able to:

- build a model of an energy-efficient house, applying their understanding of heat transfer
- explain how they have applied their understanding of heat transfer in the design of this house.
- conduct an evaluation of the efficiency of their house model.

Key vocabulary

Energy efficiency, insulation, conduction, convection

Equipment list

Each GROUP will require:

- cardboard to build basic house structure
- 50 mL beaker
- thermometer
- clear plastic sheet for windows
- tape
- tray or flat surface on which to build the model
- other materials as requested by each group and dependent on their own design.

Things to consider

Allow as much scope as possible for students to design their own models.

An important aspect of this activity is the evaluation. To make the exercise as realistic as possible, it is suggested that students test their models outside in natural conditions. The nature of the evaluation will depend on climate and weather conditions prevailing at the time. They may need to design a house which stays cool or warm.

Consider an infra-red thermometer, temperature probe attached to a datalogger or a simple electronic temperature meter sensor probe.



Teacher content information

The evaluation has a very precise specification: to keep a small beaker of water (modelling a person) at a comfortable temperature.

If students have problems with their design they may need to be reminded of all the ways in which heat transfers can occur. These include conduction, convection and exposure to direct sunlight.

The sorts of materials that students may need include insulating material (e.g. cotton wool), plant material (e.g. to model a tree shading the house).

Students complete the questions posed in the *Student Guide*, introducing them to the link between food chains and food webs.

Lesson Plan

Step 1: Discuss with the whole class what your overall design brief should be. Are you designing a house to keep warm or to keep cool? This will depend on where you live and what season it is.

Step 2: Your task is to design a one-room house model, built of cardboard. It should include features designed to maintain a comfortable temperature.

Design brief for house

- It should be at least 15-20 cm square.
- It should also contain at least two windows of clear plastic at least 2 cm square.
- It must have a roof, also of cardboard.
- It should be designed to achieve and/or maintain a comfortable temperature. This will depend on where you are doing the investigation (i.e. climate and season).

Step 3: You will evaluate how effective your design is by putting a 50 mL beaker of water inside your house and measuring its temperature before and after your test. The test will involve placing the house outside, in direct sunlight.

The beaker of water should start at a comfortable temperature, between 20 and 30 °C.

Step 4: Compare your results with the rest of your class.

Discussion

In your **Notebook** list the design features you included in your house.

Include an explanation of why you used each one.

Include a statement of how you could improve your design based on your evaluation and comparison with other designs.



6.4 Honda ad

Curriculum links

- ACSSU155 Energy appears in different forms, including movement (kinetic energy), heat and potential energy, and energy transformations and transfers cause change within systems.
KEY IDEAS
- Energy is the capacity to do work.
 - Energy transfers involve change.
 - Energy looks different in different situations – it can be transferred from one object to another.
 - Objects can have mechanical energy because of how they are moving or because of their position.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- apply many aspects of their understanding of energy forms, stores and transfers and transformations to a video involving many physical interactions.

Key vocabulary

Energy, gravitational potential energy, kinetic energy, electrical potential energy.

Things to consider

This activity could be provided to students individually as homework or used as the basis of a classroom discussion. The student record could be used as **formative assessment** of their understanding of energy before the final test.

Teacher content information

The aim of this advertisement is to give the impression of a car as an intricate and smoothly operating mechanical system. In this sense it provides a good model of how important the management of energy stores and transfers is in any mechanical system. But note that energy is actually stored in various places along this chain. There are some interesting and quite well hidden energy stores that lead to unexpected behaviour of some of the elements of the chain too. (In particular, keep an eye on the tyre that rolls up the ramp).

It took 7 months to produce this 120 second commercial. It is "regarded as one of the most ground-breaking and influential commercials of the 2000s and received more awards from the television and advertising industries than any commercial in history." (Wikipedia [https://en.wikipedia.org/wiki/Cog_\(advertisement\)](https://en.wikipedia.org/wiki/Cog_(advertisement))).

You may like to also show students the four minute video showing the making of the Honda Accord Cog Commercial (<http://wklondon.com/work/cog/>).



Lesson Plan

Step 1: Students observe the video clip.

Step 2: Students note all energy forms seen.

Step 3: Students identify where and how energy has been stored in this system (e.g. as gravitational potential energy or electrical energy).

Discussion

The **Notebook** questions could form the basis of a whole class discussion.



⚠ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

⚠ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

6.5 Sample test

A sample summative test (editable version) and marking scheme are available to teachers by email request from *Science by Doing* (sbd@science.org.au). In your email request please include your school (or university) and contact details.

Note - *Science by Doing* provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. *Science by Doing* encourages teachers to modify the items to suit individual teaching programs.



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